Sharahbill Mohamed Abdi

Data & Machine Learning Engineer · MEAL & Research

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MACHINE LEARNING / FORECASTING / MODEL DEPLOYMENT / MONITORING & EVALUATION / SURVEY RESEARCH

PROFILE

Data and machine learning engineer combining **production-oriented ML engineering** with **Monitoring, Evaluation, Accountability and Learning (MEAL)** in development programmes. At Cortech Consulting Group, I lead MEAL and research for clients including the **World Bank, IGAD, IOM, the African Union and the Federal Government of Somalia**, owning the full chain from framework design and KoboToolbox surveys to Python analysis and evaluation reporting.

I build and deploy ML systems in forecasting, classification and computer vision, shipped as live APIs and dashboards on FastAPI, Streamlit and Render, including a 1.11 MB quantized cassava-disease classifier and an XGBoost drought forecaster with 95.1% explained variance. Top 10% in a Kaggle Data Visualization Challenge, and an experienced Python data analysis trainer.

CORE SKILLS

PROGRAMMING & DATA	Python, Pandas, NumPy, SQLite, SQLAlchemy, Git & GitHub, Jupyter
MACHINE LEARNING	Classification, Regression, Time-Series Forecasting, Feature Engineering, Model Tuning · scikit-learn, XGBoost, Optuna, SHAP
DEEP LEARNING	Computer Vision, Transfer Learning · TensorFlow / Keras, TFLite quantization
DEPLOYMENT & APIS	FastAPI, REST APIs, Streamlit, Render, pytest, LLM API integration (Qwen)
MEAL & RESEARCH	MEAL Frameworks, Survey Design, Household & Community Surveys, KoboToolbox, Evaluation Reporting
ANALYSIS & VIZ	Data Cleaning, EDA, Statistical Analysis · Tableau, Plotly, Matplotlib, Seaborn · GeoPandas, OSMnx, Folium
PROFESSIONAL	CRISP-DM, Technical Communication, Training & Mentoring

LANGUAGES

Somali (Native) · **English** (Fluent) · **Kiswahili** (Fluent)

EXPERIENCE

MEAL Officer / Research

2025 – Present

Cortech Consulting Group · Nairobi, Kenya · Monitoring, Evaluation, Accountability & Learning

- Lead MEAL and research across the firm's 2025–2026 consulting portfolio for government, multilateral and NGO clients, designing and managing the MEAL frameworks that structure how each assignment is monitored, evaluated and turned into learning.
- Run field data collection on KoboToolbox across multiple surveys, including household and community surveys.
- Clean, analyse and model survey data in Python, converting raw field data into evidence, then author client evaluation and research reports that translate the analysis into clear, evidence-based findings.

SELECTED ASSIGNMENTS

2026

Ministry of Agriculture & Irrigation, Federal Government of Somalia: community early warning and disaster risk reduction (DRR) training support, including system design.

IGAD: end-of-project evaluation of "Improving Land Governance in the IGAD Region" (Phase II), covering implementation, impact, sustainability and lessons learned.

2025

World Bank: capacity assessment of Somalia's Federal Member State Ministries of Health.

IOM & African Union: monitoring and implementation framework for the Free Movement of Persons Protocol.

Heifer International Kenya: agritech landscape study on scaling digital agriculture.

Danish Refugee Council: mapping of financial service providers for refugees and host communities.

Welthungerhilfe: evaluation of "Chukua Control".

Qatar Red Crescent: SME financing capacity building in Somalia.

EXPERIENCE *cont.*

Data Analysis Trainer (Part-Time)

Trainify Institute of Technology · Nairobi, Kenya (Remote / Hybrid)

- Delivered structured Python data analysis training (Pandas, NumPy, visualisation), turning complex concepts into practical skills.
- Mentored multiple cohorts through real-world, end-to-end data projects, guiding students from raw data to presented insights.

PROJECTS

CassavaWatch · *Computer Vision, Full-Stack ML*

GitHub · Live app

- Built and deployed a mobile-first app that diagnoses cassava disease from 3–6 leaf photos and returns advice in English or Swahili, with offline queueing and a FastAPI backend running a Poisson-based outbreak detector.
- Fine-tuned MobileNetV3Small on iCassava 2019 field images to **75.2% accuracy** on a balanced 500-image test set, quantized to a **1.11 MB TFLite model** that fits a 512 MB free-tier server.

STACK Python, TensorFlow / Keras, TFLite, FastAPI, SQLite, Leaflet, Qwen API, pytest, Render

SPEI Drought Forecasting Platform · *Time-Series Forecasting*

GitHub · Live app

- Engineered a 10-stage ML pipeline forecasting SPEI-12 drought for Algeria's Oran region 1, 3 and 6 months ahead from 76 years of monthly climate data, with SHAP explainability and bootstrap prediction intervals.
- Achieved **95.1% explained variance** and **98.3% directional accuracy** with Optuna-tuned XGBoost on 28 engineered features under a strictly chronological split, shipped as an interactive Streamlit dashboard.

STACK Python, pandas, scikit-learn, XGBoost, Optuna, SHAP, Streamlit, Plotly

Nairobi Flood Guard · *Geospatial ML, Team Project*

GitHub · Live app

- Contributed to a team-built flood early-warning and matatu-rerouting system covering **1,450 Kenyan wards**, delivered through FastAPI, SMS and Streamlit.
- The team's calibrated XGBoost model reached **ROC AUC 0.889** and **recall 0.80** under county-held-out spatial cross-validation, selected over logistic regression, random forest and neural network benchmarks.

STACK Python, GeoPandas, OSMnx, XGBoost, FastAPI, Streamlit, Folium, Docker, GitHub Actions

FURTHER PROJECTS

Customer Churn Prediction — GitHub

Built a recall-optimised churn classifier on 3,333 telecom records (~15% churn): leakage-safe pipeline (stratified 75/25 split, ColumnTransformer), GridSearchCV-tuned Random Forest, and top drivers (daytime charges, 3+ service calls, international plans) in a Tableau-ready risk export.

STACK Python, pandas, scikit-learn, Tableau

NSE Stock Price Forecasting — GitHub

Forecast Safaricom (SCOM) prices from 2025 NSE data with a Random Forest on lag and calendar features to an **MAE of KES 0.60**, using a chronological 80/20 split to prevent look-ahead leakage; lag price and daily high/low range were the strongest predictors.

STACK Python, pandas, scikit-learn, Matplotlib, Seaborn

Mobile User Behaviour Classification — GitHub

Classified 700 mobile users into five behavioural tiers from device telemetry with Random Forest (1.00 accuracy and F1); app usage time (~25.6%), battery drain (22.8%) and app count (20.3%) led.

STACK Python, pandas, NumPy, scikit-learn, Seaborn, Matplotlib

EDUCATION

BSc in Mathematics and Computing
The Open University of Kenya

Data Science · Moringa School · April 2026

Data Visualization with Python · Moringa School · 2025

CERTIFICATIONS & AWARDS

Top 10%, Kaggle Data Visualization Challenge

Best Problem Solver · Moringa School Bootcamp

ICDL Certification

Nairobi Flood Guard: recognised as the best project of 2026